

Final Proposal:

Predicting Cardiovascular Disease Using Clinical Features: A Machine Learning Approach

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DSE6311 Capstone

Background and Question

Cardiovascular disease (CVD) remains the leading cause of death for men and women globally. The published research on machine-learning-based CVD prediction is well represented in literature. Most of the work stops at accuracy and AUC metrics. A model that achieves 92% accuracy but misses 40% of true positive cases would be impressive statistically but dangerous clinically. What do model errors mean in a clinical context? This project attempts to answer that question by reframing model evaluation as a clinical risk question as opposed to a purely statistical one. A recent review of ML studies for the prediction of heart disease supports this by finding that model selection should be based on performance metrics crucial for clinical diagnostic accuracy rather than overall accuracy alone (Kumar et al.,2025).

The focus will not be on whether a model can predict CVD, but on whether performance metrics can be translated into meaningful clinical risk assessment. How should model performance be evaluated, given the trade-off of sensitivity and specificity, to assess clinically appropriate risk in the context of cardiovascular disease?

Platforms that detect coronary artery disease lesions using AI-driven diagnostic tools are becoming standard in clinical imaging environments. The ability to thoroughly assess model performance against clinical risk thresholds is a critical skill. Validation teams must ask themselves not only if a model is accurate, but also if the model error profile can be clinically trusted and safe for patient outcomes. This project will simulate the validation process to make it both academically rigorous and professionally meaningful.

Cardiovascular disease prediction using the UCI Heart Disease dataset is well used in machine learning publications. This project is not novel in that sense. The project will aim to reframe and improve the approach from a clinical validation standpoint and apply it to model evaluation. Academic projects treat model evaluation as a report of sensitivity and specificity metrics. Clinical AI development treats those same metrics as design decisions with real consequences for patient safety. This project applies that clinical standard to an academic context.

The primary stakeholder identified is AI/ML developers and clinical validation teams working in medical imaging and diagnostic AI. They are the ones responsible for assessing whether a model is safe for deployment in a clinical workflow. Secondary stakeholders include cardiologists and clinical informaticists who use and interpret these tools for patient point of care.

Hypothesis and Prediction

Hypothesis: Optimizing a machine learning model's sensitivity will reduce false negatives but increase false positives. In cardiovascular disease prediction, this trade-off carries measurable clinical complications. A model cannot be evaluated as clinically appropriate without also analyzing what its error profile means in a real-world care setting. It is also important to acknowledge that false positives are not a trivial outcome. Overcalling healthy patients increases patient anxiety, drives up healthcare costs, and erodes patient trust in healthcare. The goal is not

to minimize false negatives by any means necessary, but find the threshold where the tradeoff is clinically defensible.

Prediction: A tuned ensemble model (Random Forest or XGBoost) optimized for sensitivity will outperform a baseline Logistic Regression model on recall and ROC-AUC. However, the most clinically appropriate model will not necessarily be the highest performing model based on accuracy, it will be the model whose sensitivity (true positive)/specificity (true negative) balance best aligns with defined clinical risk thresholds. Cost-sensitive learning will also be explored as a potential approach. CSL assigns a higher misclassification penalty to false negatives during model training to reflect the clinical reality that missing a true positive carries greater risk than a false negative (Brownlee, 2020). Sensitivity will be prioritized as the primary performance target, as false negatives could have dire patient outcomes, more so than false positives. Sensitivity and specificity threshold goals are $\geq 85\text{-}90\%$ and $\geq 75\text{-}80\%$, respectively. These thresholds are chosen based on literature findings in a 2024 clinical feature analysis for CVD risk screening using XGBoost. A sensitivity of 89.1% and specificity of 85.39% were achieved and selected using the Matthews Correlation Coefficient, while keeping false negative rate clinically acceptable (Vyshnya et al, 2024).

Foundational clinical methods literature establishes that for cardiovascular diagnostic screening, a sensitivity of 75% and specificity of 80% represent a reasonable clinical benchmark. These metrics must always be interpreted together with disease prevalence and clinical context rather than isolation (Simon and Boring, 1990).

These thresholds reflect the clinical reality that a missed cardiovascular diagnosis carries greater risk than an unnecessary follow-up.

Dataset and Variables

The UCI Heart Disease dataset is one of the most widely used datasets in cardiovascular machine learning research. The dataset was originally collected across four institutions. The original dataset contains 76 attributes, although all published experiments have referred to a subset of 14 of them. The Cleveland database is the only one that has been used by machine learning researchers to date. The experiments have focused on distinguishing between the presence of disease from absence.

The dataset is formally cited as: Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart Disease [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C52P4X>.

The version used for this project is the Cleveland dataset and may be accessed here:

[Heart-UCI-Dataset/heart.csv at master · sharmaroshan/Heart-UCI-Dataset · GitHub](#)

The dataset includes 303 patient records with 13 clinical and diagnostic features commonly used in CVD risk assessment. A binary target variable is included, indicating disease presence. The dataset is a well-known and credible source, however, it is aging. The goal of the project is not to deploy a clinical tool, but to simulate and demonstrate the clinical validation process that would

be applied to one.

Response / Outcome Variable

Target (binary): 0 = No cardiovascular disease, 1 = Cardiovascular disease present.

Predictor Variables

All 13 clinical features will be used as candidate predictors: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, ST slope, number of major vessels, and thalassemia type. Feature importance and ranking will be performed during pre-processing and model development to identify which clinical variables carry the most predictive weight.

Tentative Analysis Plan:

This project will include both supervised and unsupervised learning components and be structured to resemble a clinical AI validation workflow.

Unsupervised – Dimensionality Reduction:

Principal Component Analysis (PCA) will be used to decide if dimensionality reduction is necessary due to potential multicollinearity among the 13 features. This decision to reduce will be based on whether PCA improves the analysis or creates new problems. The goal of this project is accurate clinical risk assessment and not feature interpretability, so uninterpretable features are not a concern here. This component of the plan may be refined as the project develops.

Supervised – Classification:

Three classification models will be built and compared for supervised learning and used to predict the binary outcome of cardiovascular disease presence. Logistic Regression as a baseline, and Random Forest and XGBoost as tunable ensemble methods. Both will go through cross-validation and hyperparameter tuning.

Logistic Regression – This will serve as the baseline model. Logistic Regression is clinically relevant and will provide well understood coefficients to secondary stakeholders.

Random Forest and/or XGBoost – These will serve as the tunable ensemble methods for this project, and can be used for cross-validation and hyperparameter optimization. A JMIR study comparing Random Forest, XGBoost, Logistic Regression, and other models for coronary heart disease prediction found that Random Forest was the best performing model due to its sensitivity/specificity, and AUC (Vu et al., 2025). These models will be evaluated for performance over baseline, as well as if those gains come with an effect on interpretability and clinical explanation.

Caveat to chosen models and JMIR study – This study applied a similar set of supervised machine learning models to a longitudinal Japanese cohort for coronary heart disease prediction. It also used comparable evaluation methods including sensitivity, specificity, AUC, and decision curve analysis (DCA). I want to acknowledge the methodological overlap, and highlight how this project differs in many ways. The JMIR study was designed to identify novel population specific risk factors using 7672 patients followed over 15 years and is written for clinical researchers. This project is using the UCI Heart Disease dataset as a proof-of-concept environment, with the goal of demonstrating how a clinical validation team would assess a decision support tool before deployment. Next, this project will include cost-sensitive learning, comparing standard and cost-penalized versions of the same models to isolate the effect of including clinical risk into the training itself. And, while both projects incorporate DCA, in this project it will serve as part of the central argument, that model selection in a clinical setting should be driven by benefit across the decision thresholds, and not just accuracy. I feel that the existence of comparable methodology in peer-reviewed publications supports the validity of this approach rather than a reduction in the novelty of the project.

Evaluation:

Models will be evaluated on accuracy, precision, recall, specificity, F1-score, and ROC-AUC. A threshold analysis will also be run to see how shifting the classification cutoff affects the sensitivity/specificity tradeoff and what that means from a clinical standpoint.

To account for the small sample size, stratified k-fold cross-validation will run across all model training and evaluation steps. Keeping the class ratio consistent across folds is important given the small and imbalanced dataset. Regularization is built into all three models and will be monitored during hyperparameter tuning to reduce overfitting risk.

A Decision Curve analysis (DCA) will be applied to assess the clinical usefulness of each model across a range of decision thresholds. This method was introduced by Vickers and Elkin (2006) as a framework designed to evaluate whether a prediction model is clinically useful, and not just statistically accurate.

The central question for evaluation will be which models' error profile is most appropriate in a clinical decision support context, which is what must be answered before deployment.

Potential Pitfalls:

- Small sample size (n=303): This limits how well results will generalize and will be noted as a key limitation. Since the project is not making population-level claims, the actual risk is that 303 samples may not be enough to provide a robust model. Steps for cross-validation outlined in the evaluation section will attempt to mitigate this concern.

- This dataset is known to have a class imbalance. If one outcome is more dominant than the other, then accuracy will be a poor metric. Sensitivity, specificity, and ROC-AUC will be used instead. Oversampling techniques such as SMOTE will be explored as a proof-of-concept adjustment to address the imbalance during training.
- Dataset age: The UCI dataset is an older dataset (1989). It may not reflect current patient populations. Findings will be scoped accordingly.
- Stratified cross-validation, regularization, and tuning will be used throughout to account for overfitting on smaller datasets.
- Reducing features to principal components makes individual variables harder to interpret clinically. Since the goal is not interpretability, this is not a meaningful limitation here. This tradeoff will be addressed in the results for completeness.

How Will I Know if the Question Is Answered?

I will compare all three models once they have been trained and evaluated. I will compare their performance across clinical metrics and make a recommendation on which best fits a clinical decision support setting. The answer will be framed around what the results mean for a patient population, not just a report of the numbers.

How Will I Know if the Hypothesis Is Supported?

The hypothesis will be supported if there is an observed trade-off between sensitivity and specificity across the models. Tuning and cost-sensitive learning for sensitivity shifts the false negative/positive balance in a meaningful way. Model robustness will be the biggest challenge for this project. 303 data samples will need to produce models robust enough to reach sensitivity and specificity benchmarks to allow for a meaningful clinical risk assessment. If the models fall short of those thresholds, that finding is also worth discussing. In that context, the discussion moves to a focus on data requirements needed to validate clinical decision support tools in practice.

Technical Details:

Programming language used will be Python, used in Google Colab.

Additional resources used within Python will be scikit-learn, pandas, NumPy, matplotlib, seaborn, XGBoost.

GitHub Repository for this project can be found here:

https://github.com/mnooney/CVD_Capstone

References

Kumar R, Garg S, Kaur R, Johar MGM, Singh S, Menon SV, Kumar P, Hadi AM, Hasson SA, Lozanović J. A comprehensive review of machine learning for heart disease prediction: challenges, trends, ethical considerations, and future directions. *Front Artif Intell*. 2025 May 13;8:1583459. doi: 10.3389/frai.2025.1583459. PMID: 40433606; PMCID: PMC12106346.

S. Vyshnya, R. Epperson, F. Giuste, W. Shi, A. Hornback and M. D. Wang, "Optimized Clinical Feature Analysis for Improved Cardiovascular Disease Risk Screening," in *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 5, pp. 816-827, 2024, doi: 10.1109/OJEMB.2023.3347479.

Simon D, Boring JR III. Sensitivity, Specificity, and Predictive Value. In: Walker HK, Hall WD, Hurst JW, editors. *Clinical Methods: The History, Physical, and Laboratory Examinations*. 3rd edition. Boston: Butterworths; 1990. Chapter 6. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK383/>

Brownlee, J. (2020, January 14). Cost-sensitive learning for imbalanced classification. Machine Learning Mastery.
[Cost-Sensitive Learning for Imbalanced Classification - MachineLearningMastery.com](https://machinelearningmastery.com/cost-sensitive-learning-for-imbalanced-classification/)

Vu T, Kokubo Y, Inoue M, Yamamoto M, Mohsen A, Martin-Morales A, Dawadi R, Inoue T, Tay J, Yoshizaki M, Watanabe N, Kuriya Y, Matsumoto C, Arafa A, Nakao Y, Kato Y, Teramoto M, Araki M, *Machine Learning Model for Predicting Coronary Heart Disease Risk: Development and Validation Using Insights From a Japanese Population-Based Study*
JMIR Cardio 2025;9:e68066
URL: <https://cardio.jmir.org/2025/1/e68066>
DOI: 10.2196/68066

Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making*. 2006 Nov-Dec;26(6):565-74. doi: 10.1177/0272989X06295361. PMID: 17099194; PMCID: PMC2577036.

Data Dictionary

age: age in years

sex: gender (1 = male, 0 = female)

cp: chest pain experienced (1: typical angina, 2: atypical angina, 3: non-anginal pain, 4: asymptomatic)

trestbps: resting blood pressure (mm Hg on hospital admission)

chol: serum cholesterol measurement in mg/dl

fbs: fasting blood sugar (> 120 mg/dl, 1 = true; 0 = false)

restecg: Resting ECG measurement (0: normal, 1: having ST-T wave abnormality, 2: probable or definite left ventricular hypertrophy by Estes' criteria)

thalach: maximum heart rate achieved

exang: Exercise induced angina (1 = yes; 0 = no)

oldpeak: ST depression induced by exercise relative to rest ('ST' refers to ECG waveform)

slope: the slope of the peak exercise ST segment 1: upsloping, 2: flat, 3: downsloping)

ca: The number of major vessels 0-3 colored by fluoroscopy

thal: A blood disorder called thalassemia (3 = normal; 6 = fixed defect; 7 = reversable defect)

target: Heart disease (0 = no, 1 = yes)

Source: [Heart Disease - UCI Machine Learning Repository](#)

[Heart-UCI-Dataset/heart.csv at master · sharmaroshan/Heart-UCI-Dataset · GitHub](#)